

Anish Rao Toorpu

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EDUCATION

North Carolina State University, Raleigh NC, USA

August 2023 - May 2025

Master of **Computer Science**

GPA: 4.0/4.0

Coursework: Neural Networks and Deep Learning, Design and Analysis of Algorithms, Software Engineering, Automated Learning and Data Analysis, Object-Oriented Design & Development, User Experience, Operating systems.

Anurag University, Hyderabad, India

August 2019 - May 2023

Bachelor of Technology in **Artificial Intelligence**

GPA: 8.66/10.0

Coursework: Data Structures & Algorithms, Operating Systems, Object-Oriented Programming, Database Management System, Software Engineering, Machine Learning, Deep Learning, Neural Networks, Natural Language Processing.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, TypeScript, SQL

Web Technologies: HTML, CSS, React, Node.js, REST APIs, GraphQL, AWS, GCP

Databases: MySQL, Oracle, Redis, PostgreSQL

Tools/ Frameworks: Git, Ruby on Rail, Spring Boot, Kafka, Kubernetes, Jira

Others: R-Studio, Linux, HBase, Docker, CI/CD, DevOps

EXPERIENCE

Software Engineer Intern, iSimcha, LLC

May 2025 - Present

- Developed a **React (TSX)** front end paired with **Python** functions for **LLM**-powered summary generation, boosting end-to-end processing throughput by 30%.
- Fine-tuned domain-specific **LLM's** on healthcare datasets reducing model **perplexity** by **18%** and improving translation accuracy for complex medical terminology and integrated **LiteLLM** for inference calls.
- Wrote and deployed **Google Cloud Functions** on **Firestore** to orchestrate model workflows and store results in **Firebase**, and collaborated on evaluation protocols to improve model selection efficiency by 25%.

Software Engineer, North Carolina State University

September 2023 - May 2025

- Developed and implemented an **LSTM-based Anomaly Detection System** for the BTEC Water Treatment Facility, improving operational efficiency by 40% through accurate forecasting and proactive issue detection.
- Engineered secure token retrieval and enhanced **GraphQL** request handling, improving data security and reducing data retrieval issues by **30%**.
- Designed visualization tools and automatic email notifications, facilitating proactive operational management and timely responses to anomalies, resulting in a **25%** improvement in system reliability.
- Created and managed a Python backend on AWS (**Flask, pandas, S3 & RDS**) to automate student mental-health and transfer-readiness reports, cutting manual work and boosting accuracy.
- Built serverless data APIs with **AWS Lambda** and API Gateway to serve processed datasets with low latency, automatic scaling, and integrated IAM-based access control, simplifying consumption by analytics and reporting tools.
- Set up **AWS CloudFormation** and a **Docker**-based CI/CD pipeline (AWS CodePipeline) to automate deployments and keep environments consistent

Software Engineer, Cognizant Technology Solutions

February 2023 - June 2023

- Engineered and deployed scalable loan management systems using **Java** and **Spring Boot**, driving a 30% increase in operational efficiency.
- Collaborated with cross-functional teams using **Git** and **Jenkins** to identify and resolve bugs in the loan management applications, resulting in a 20% reduction in customer complaints and improved system stability.
- Automated critical processes with **Kafka** for real-time event streaming, minimizing manual efforts and boosting productivity while achieving notable cost reductions.

PROJECTS

Event Management System (Ruby on Rails)

- Developed a comprehensive event management system using **Ruby on Rails**, allowing users to manage events, book tickets and leave reviews.
- Executed functionalities for both admin and attendees, including event creation, ticket booking, attendee management, and review systems, resulting in a **25%** increase in user engagement and satisfaction.
- Integrated bonus features such as filtering attendees by event name and allowing users to book tickets for other users.

Data Query System (Spring Boot, React.js, MySQL, OAuth 2.0 & JWT)

- Spearheaded the development of a streamlined data query system, facilitating efficient access to student information from existing event records.
- Implemented a robust solution utilizing **Spring Boot** for backend operations and **React.js** for frontend interactions, seamlessly integrating a **MySQL** database for efficient management and analysis of student and event data.
- Integrated **OAuth 2.0** and **JWT** (JSON Web Tokens) for secure user authentication and authorization, enabling secure access across multiple platforms while protecting sensitive data.